FIT PM12
Hardware test dossier

LEGACY-SN1803-UNBOUND

HISTORICAL — full dossier

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INCOMPLETE / NOT RELEASED

Historical multi-date engineering evidence. Factory DNA was not recorded during these measurements. This is not a new batch acceptance certificate.

Prepared by _____

Reviewed by _____

Disposition / deviations _____

Signature and date _____

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1 Scope, identity and disposition

This dossier records evidence and test coverage. A long report is not proof of compliance: incomplete channels, unknown configuration, absent raw data and failed restoration remain visible.

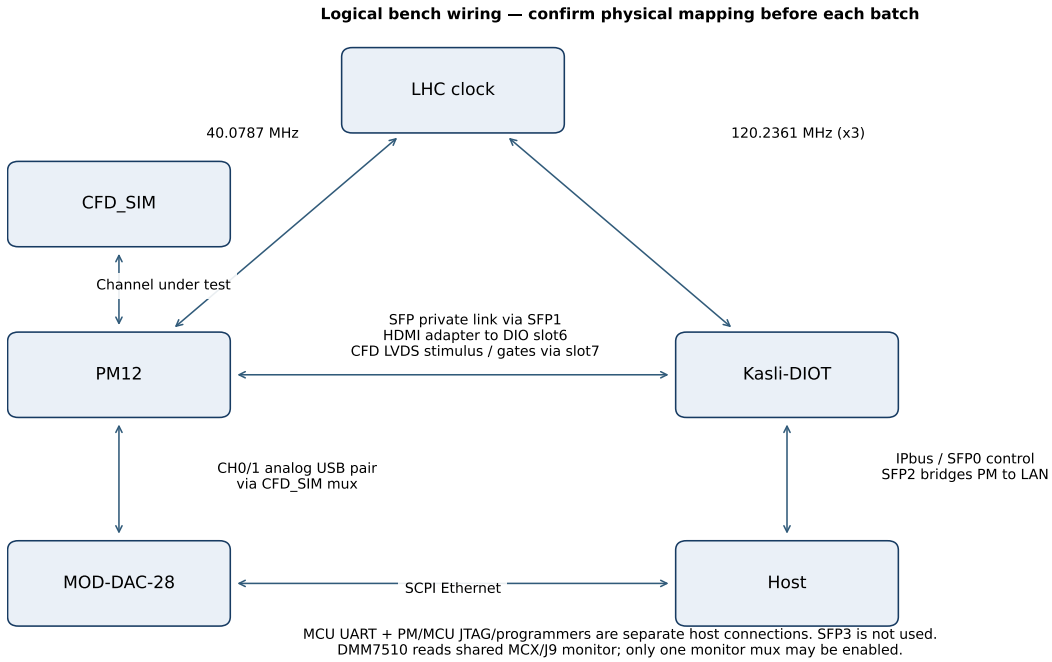
Measurements from multiple dates/configurations. No factory DNA was recorded at acquisition. Not one acceptance run and not automatically bound to a newly enrolled board.

Field	Recorded value
Hardware identity	LEGACY-SN1803-UNBOUND
MCU EEPROM label	1803
Evidence mode	HISTORICAL
Merged batches	1
Analog channels measured	12
Final acceptance	NOT RELEASED — requires complete required coverage, frozen limits and reviewer disposition
Missing channels	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Canonical new board label: PM12-K7-<full FPGA factory DNA>. Preserve leading zeroes. JTAG IDCODE identifies chip type, not the individual device. EEPROM serial and debugger cable serial are secondary labels. Replacing the FPGA changes this hardware identity and requires an explicit board genealogy record.

Historical evidence predating factory-DNA reads is never assigned to a newly enrolled DNA automatically. No chip or EEPROM is programmed to assign this number; the printed asset label and registry use the hardware value.

2 Bench inventory and wiring



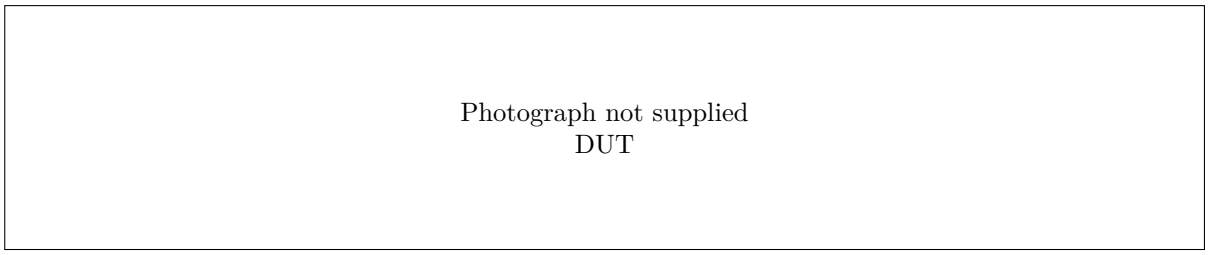
Logical wiring, not a verified pin-level schematic. Physical port/channel mappings belong to each acquisition snapshot.

Connection	Physical assignment / observation
PM data LVDS via HDMI	Kasli slot6 channels0..3 are inputs; observed PM lane order3,1,2,0
Shared HDMI SPI	Slot6:SELECT4,CLOCK5,MISO6,MOSI7; only4/5/7 may drive; mask0x00b0
CFD test and phase observation	Slot7 channel0:CFD stimulus; phase RX8/9 correspond to slot7 channels1/3
PM SFP	Private cable to Kasli SFP1, line rate1.202361Gb/s; not standard switch-facing1000BASE-X
Kasli Ethernet	SFP0:control IPbus; SFP2:optical PM bridge to LAN; SFP3 unused
Common reference	PM40.0787MHz; Kasli120.2361MHz; stated generator settings, no frequency-counter calibration

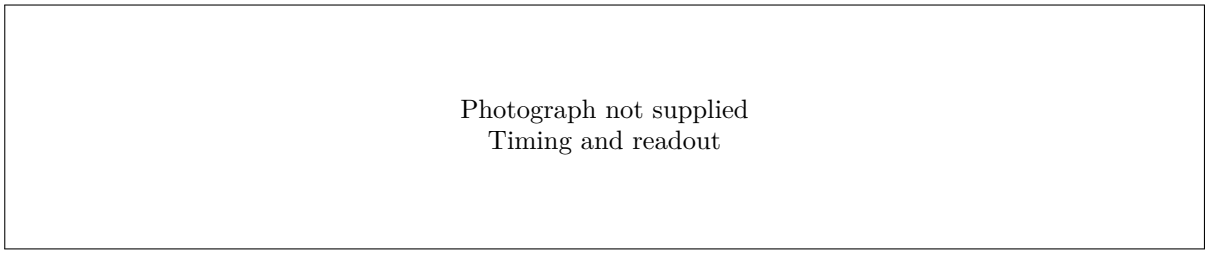
Role	Module	Identity
DUT	FIT PM12	1803
Timing and readout	Kasli-DIOT	00001
LVDS stimulus/capture	DIO-LVDS RJ45 slot 6	NOT READ
LVDS stimulus/capture	DIO-LVDS RJ45 slot 7	NOT READ
Analog stimulus	MOD-DAC-28	ESP32-MAC-6825DD4C49E4
DC voltage measurement	Keithley DMM7510	04679690
Analog adapter	CFD_SIM channel 12	NOT READ
Clock reference	External LHC generator	NOT READ

Photographs supplement, but never replace, hardware identity. Each photo is associated with the identified module and its file hash. Missing photos or unread device IDs are left explicit.

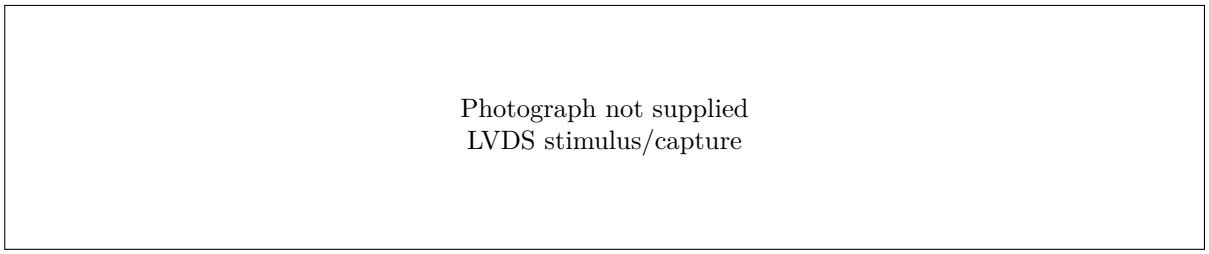
FIT PM12



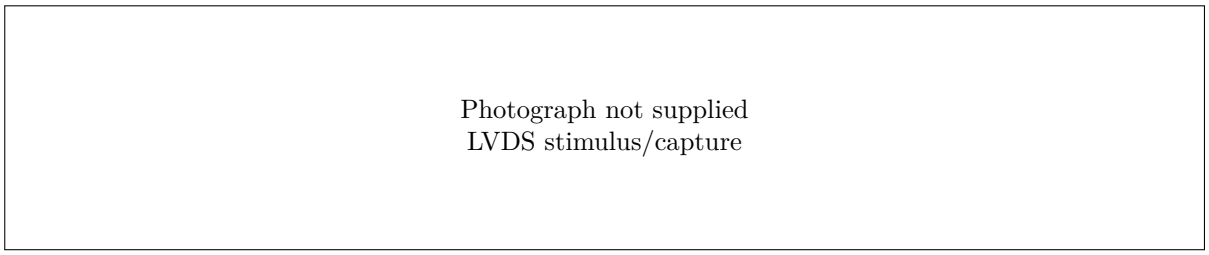
Kasli-DIOT



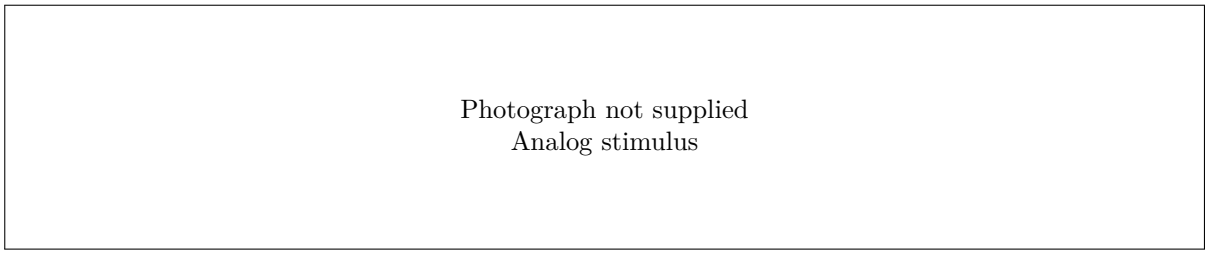
DIO-LVDS RJ45 slot 6



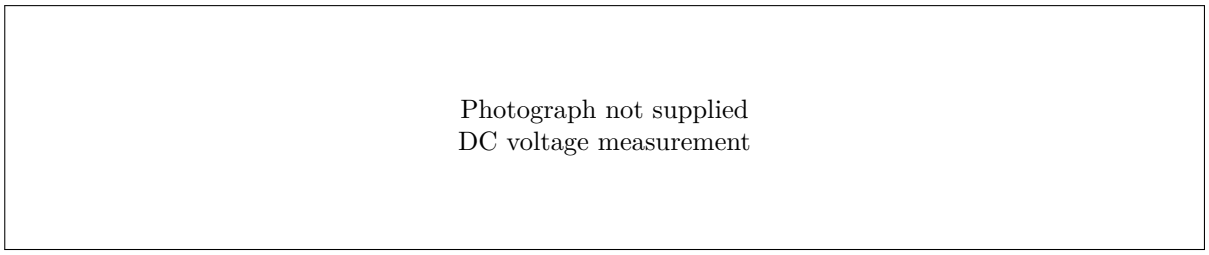
DIO-LVDS RJ45 slot 7



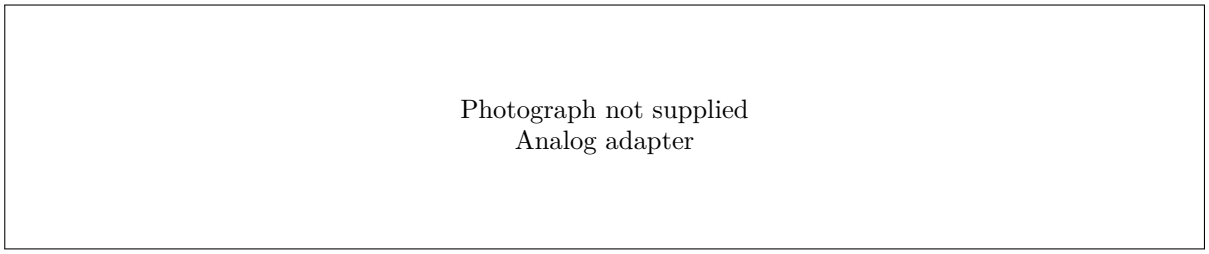
MOD-DAC-28



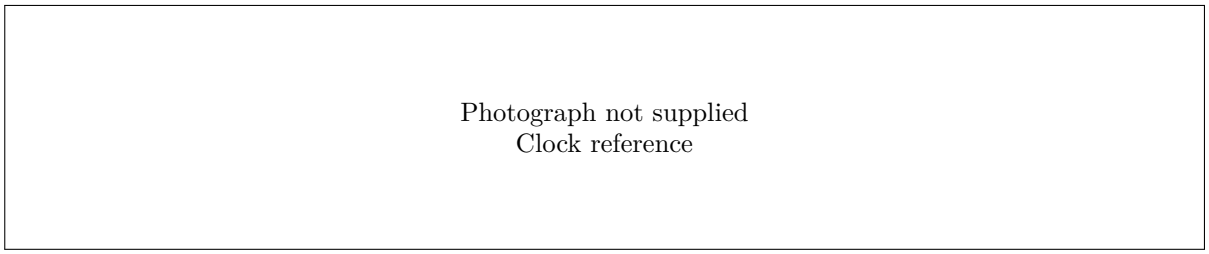
Keithley DMM7510



CFD_SIM channel 12



External LHC generator



3 Test coverage and acceptance rules

Test	Result	Scope / deviation
JTAG chain / 168 static links	HISTORICAL PASS	2026-10-13; exact reduced measured logs retained
CPU–FPGA SPI / peripherals	HISTORICAL PASS	2026-10-15; original diagnostic image
Configuration flash read / ownership	HISTORICAL PASS	Read-only ownership sequence; stored-image boot NOT CONFIRMED
Power startup / timeout fault injection	NOT RUN	Static PGOOD cannot substitute for fault testing
GTX PRBS / Ethernet loopback	HISTORICAL PASS	Old exact 40 MHz setup, not current LHC private link
I2C mux / power monitors	HISTORICAL PASS	2026-10-14; uncalibrated currents, missing P3V8 shunt
HDMI / digital ADC / TDC	HISTORICAL PASS	2026-11-06; all digital ADC channels, stimulated TDC only channel 12
CFD monitor and B/C/D DAC curves	HISTORICAL	Channel 12 only; restored-code correction retained
Gate phase and DAC-A	HISTORICAL PASS	Channel 12,20 settings; nominal sample time units
ADC transfer / adjacent bits	HISTORICAL PASS	2026-11-07 channel 12, A/B; 257 levels, raw revalidated

PASS requires the configured process to finish successfully AND every declared result check to match. NOT_RUN, BLOCKED, ERROR and INCONCLUSIVE cannot become PASS. Failed cleanup is a failure. Repeated runs do not overwrite earlier evidence.

4 Test procedure and criteria

4.1 Hardware identity and PGOOD

Read factory FPGA DNA and MCU EEPROM serial under leases. Match enrolled UID, integrated build and stable physical PGOOD.

Limits: MCU EEPROM serial and FPGA IDCODE are not unique factory identity. CPU factory signature not read.

4.2 JTAG chain and168 static CPLD–FPGA links

Validate13 IDCODEs. Program checksum-verified link diagnostic into FPGA SRAM only.338 walking-one/zero vectors in each of gateware and independent boundary-scan receive paths. Restore checksum-verified integrated image, initialize private link, verify build and repeat HDMI/ADC/TDC functional tests.

Limits: Static connectivity, not at-speed margin or analog quality. Expected SYS reset from deliberate FPGA SRAM reconfiguration; never reset or program MCU/CPLD/flash.

4.3 GTX internal PRBS, MAC loopback and external link

Exercise GTX PRBS and MAC loopback with negative controls; restore integrated image. External PM↔Kasli private link checked separately by integrated suite.

Limits: Legacy1000BASE-X image expects old exact40MHz clock. Requalify with present40.0787MHz; old PASS is historical only.

4.4 I2C mux address inventory and bus topology

Single-branch scans through both muxes, address readback and decoded monitor IDs; preserve current/voltage samples.

Limits: Legacy runner programs older FPGA and restores obsolete baseline. MCU telemetry is integrated, exhaustive topology adapter still requires migration.

4.5 FPGA configuration flash ownership and read-only access

Verify ownership guard, PROGRAM_B hold, JEDEC and bounded READ03, bus release, reload integrated SRAM image.

Limits: Destructive to volatile FPGA state. Qualified current-image restoration/recovery adapter required; stored-image boot remains unconfirmed.

4.6 Power sequencing and timeout shutdown

Observe EN-to-PG rise and timeout shutdown without replacing original MCU protection; record rails and fault state.

Limits: Fault injection and power-cycle qualification NOT_RUN; static PGOOD is not equivalent.

4.7 CPU peripherals, SPI walking patterns and power telemetry

Read PING, GPIO, temperature ID, three TDC chip IDs, PLL, FPGA signature, 32 walking patterns with scratch restoration, INA and POWER. Recheck PGOOD before sensor access.

Limits: No power cycling or timeout fault injection. Unfitted P3V8 shunt: voltage only; no current acceptance. Sensor numeric limits require a frozen acceptance specification.

4.8 HDMI, all digital ADC paths and stimulated TDC

Four HDMI lanes: walking discovery and slow/fast PRBS, empty frames; SPI signature reads only. Capture both ADC banks, synthetic and physical, all12 channels. Quiet/stimulated/quiet TDC:64 pulses to configured channel. Check uptime.

Limits: TDC stimulus only on the physically connected channel. No calibrated ADC analog acceptance for unconnected channels. HDMI SPI writes disabled: bus shared with MCU.

4.9 Gate delay versus internal DAC-A

Establish explicit DAC-A0000 baseline;20 code points0..65535; common trigger and ISERDES phase buffers; restore0000 and verify phase return. Save every raw capture.

Limits: Sampled centroid, nominal sample metadata; no calibrated absolute phase/jitter claim. Original20MHz CPLD gate divider required.

4.10 ADC differential transfer and adjacent data-bit coverage

SCPI DAC differential-1.2..+1.2V, commonmode1V,257 points, ADC A/B. Decode4096-word buffers at STR edges, verify tags/generations;12 bits and11 neighbor pairs, all4 states with sufficient population. Restore1V/1V.

Limits: Requested voltage, no calibrated gain/DNL. Per-pin0..3.3V ceiling. Functional persistent-tie test, not resistance/pin-order/timing-margin proof.

4.11 CFD analog monitor DC levels

Read all8 monitor nodes with DMM7510, correct divider ratios, leave all shared monitor muxes inhibited.

Limits: Original script is channel12-specific. Parameterized multi-module inhibition and DMM/wiring qualification required.

4.12 Internal DAC B/C/D transfer curves

65 code settings perDAC, three DMM readings per setting, corrected analog-node voltage and verified explicit-code restoration.

Limits: Original script is channel12-specific. Per-channel restore codes, shared mux inhibition and DMM wiring must be configured.

5 Digital link and peripheral evidence

5.1 MCU / SPI

Recorded field	Result
scope	Physical MCU SPI with original slave protocol, passive FPGA I2C pins
rows	18 entries; see archived JSON/raw files
mcu_source	f9869c9
fpga_magic	504d5a02
nonvolatile_writes	False
fpga_load	PASS
spi_test	PASS10 runs,320 walking-bit patterns with readback and scratch restoration
restored_image	Matching passive SPI diagnostic left in SRAM

5.2 Flash ownership

Recorded field	Result
scope	Physical config bus ownership/read-only flash/reset-release; no flash write/erase
rows	18 entries; see archived JSON/raw files
jedec_id	016017
flash_ownership_read_release	PASS
boot_done_within2s	False
boot_validation	DONE did not assert; stored image unknown, not a successful FPGA boot

5.3 Ethernet / PRBS

Recorded field	Result
board	PM12
serial	210249A73E43
external_sfp	NOT_RUN
persistent_writes	False
status	PASS
bitstream_sha256	fb6e39d8375d46dd01fde00d22a6075bf78756666cb79a8cc96f235622a5
restore_bitstream_sha256	c9fbd02768ce12750dc68b8bb24c34403027a65f9e002661e29bb3d2e2d
samples	12 entries; see archived JSON/raw files
checks.initial_mac_clocks	True
checks.initial_mac	True
checks.payload_recovery_clocks	True
checks.payload_recovery	True
checks.prbs_clocks	True
checks.prbs	True
checks.prbs_recovery_clocks	True
checks.prbs_recovery	True
checks.final_mac_clocks	True
checks.final_mac	True
checks.payload_negative_control	True
checks.prbs_negative_control	True
windows.initial_mac.tick	360504425
windows.initial_mac.refcount	300420354

Recorded field	Result
windows.initial_mac.txframes	88576
windows.initial_mac.rxframes	88576
windows.initial_mac.badframes	0
windows.initial_mac.prbs_error_samples	0
windows.initial_mac.rxbytes	5314560
windows.initial_mac.sys_mhz	120.00813082556591

5.4 I2C topology

Recorded field	Result
date	2026-10-14
digital_pass	True
physical_restore_verified	True
fpga_sha256	b26bdccb44ff0d323a58637efd3cee59a98752afdf432d392f4aa47a1a3d9
nonvolatile_writes	False
mux_addresses	['0x70', '0x71']
mux_selection_readback_cases	16
reset_test	True
automatic_line_release_test	True
downstream_scan_address_range	0x08..0x77 excluding upstream mux addresses
confirmed_monitors	3
afe_device_ack_count	0
warnings	2 entries; see archived JSON/raw files
coverage_limits	['AFE routes yielded no downstream ACK; physical connectivity of unloaded branches is not proven', 'No sensor configuration or power-controller writes', 'Voltage/current calibration not validated', 'Slow host-driven I2C over UDP; high-speed I2C electrical margins not measured']
bus_voltages_by_monitor.4	[1.0, 1.2, 1.792]
bus_voltages_by_monitor.5	[1.456, 3.336, 1.784]
bus_voltages_by_monitor.6	[1.008, 5.5040000000000004, 0.0]

5.5 Power rail observations

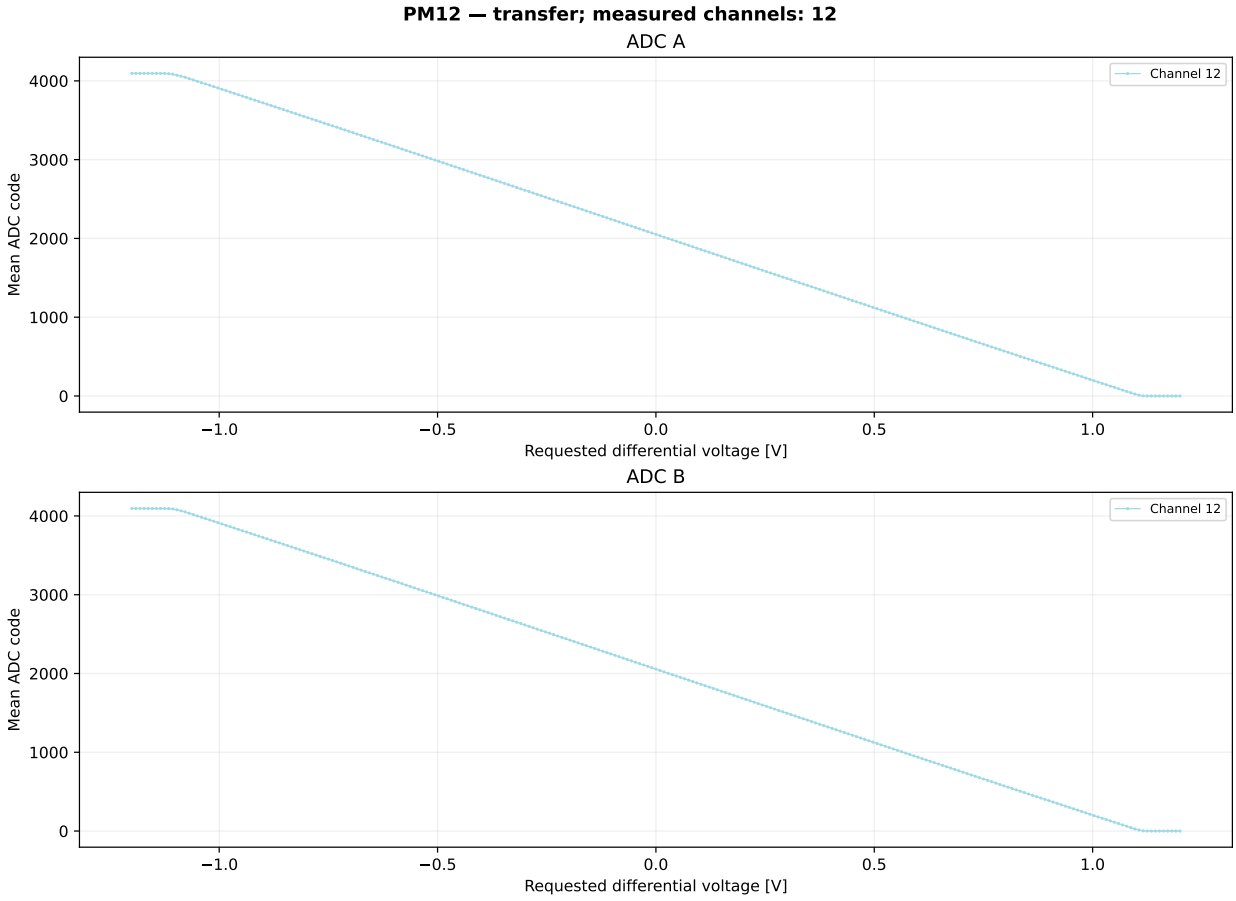
Rail	Bus [V]	Current [A]	Note
P1V0_LDO	1.0	0.1276	Uncalibrated; inspect shunt/clipping metadata
P1V2_LDO	1.2	0.092	Uncalibrated; inspect shunt/clipping metadata
P1V8_LDO	1.792	0.004	Uncalibrated; inspect shunt/clipping metadata
P3V8	1.48	None	Uncalibrated; inspect shunt/clipping metadata
P3V3	3.336	4.920000000000001	Uncalibrated; inspect shunt/clipping metadata
P1V8	1.784	0.36000000000000004	Uncalibrated; inspect shunt/clipping metadata
P1V0	1.008	0.10800000000000001	Uncalibrated; inspect shunt/clipping metadata
P5V5	5.5040000000000004	0.08	Uncalibrated; inspect shunt/clipping metadata

Rail	Bus [V]	Current [A]	Note
TP1/TP2	0.0	None	Uncalibrated; inspect shunt/clipping metadata

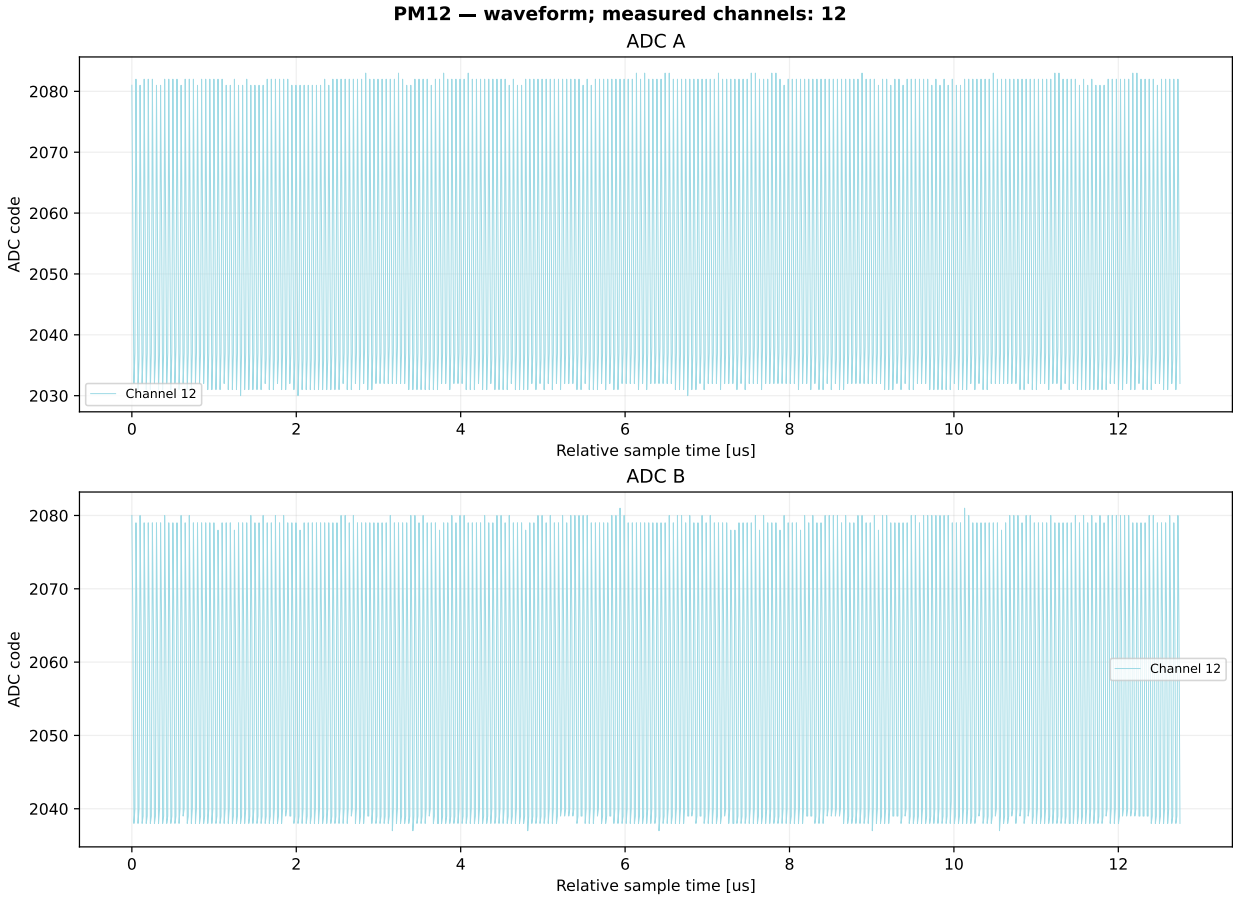
P3V8 has no fitted shunt: current is not an accepted measurement. Original clipping observations are preserved in raw telemetry and are not interpreted as a rail current.

6 CFD and ADC — channel comparison

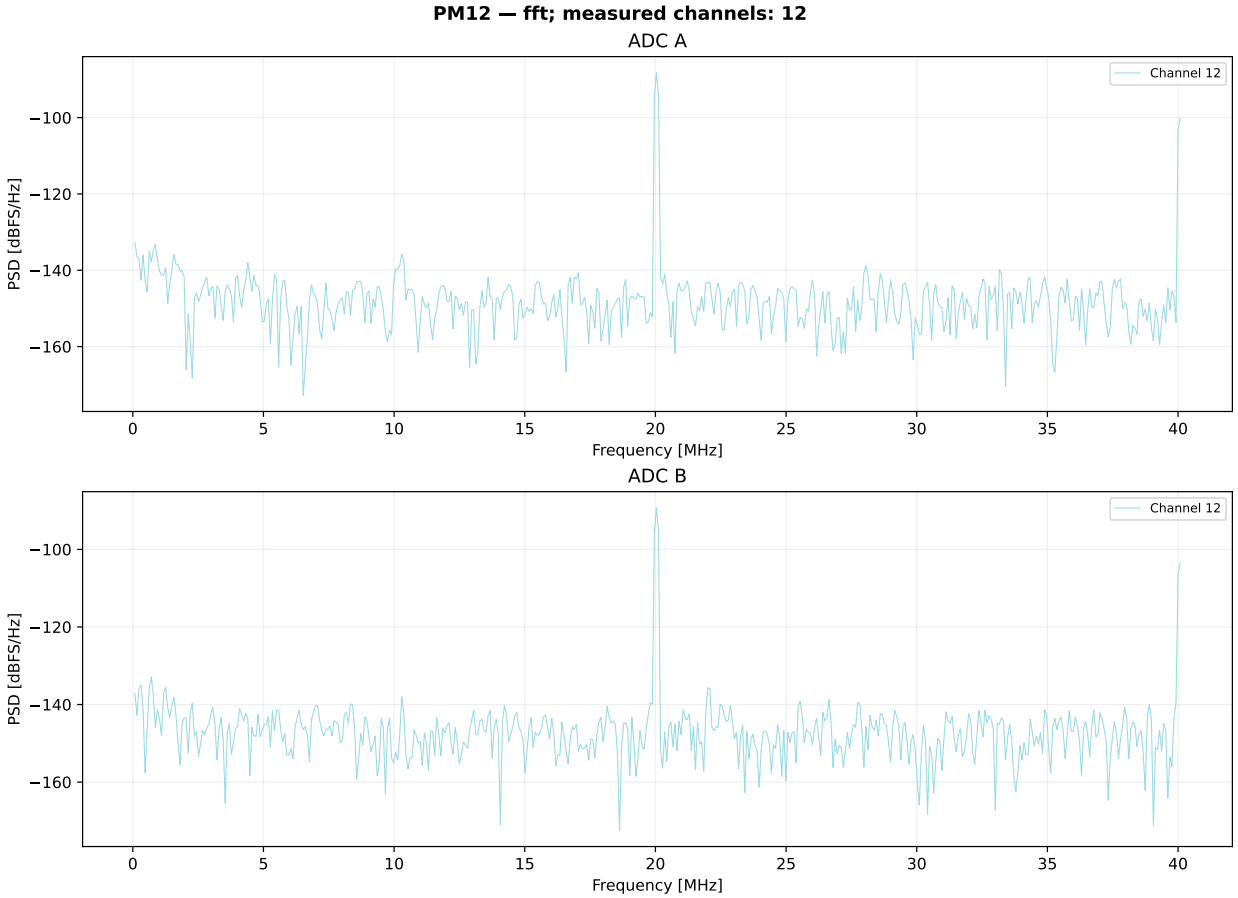
All measured channels share one overlay per observable, with a fixed channel color and common axes. ADC A/B occupy separate panels. No line is invented for an unmeasured channel. The twelve-channel requirement is independent of the number of connected simulators in a batch.



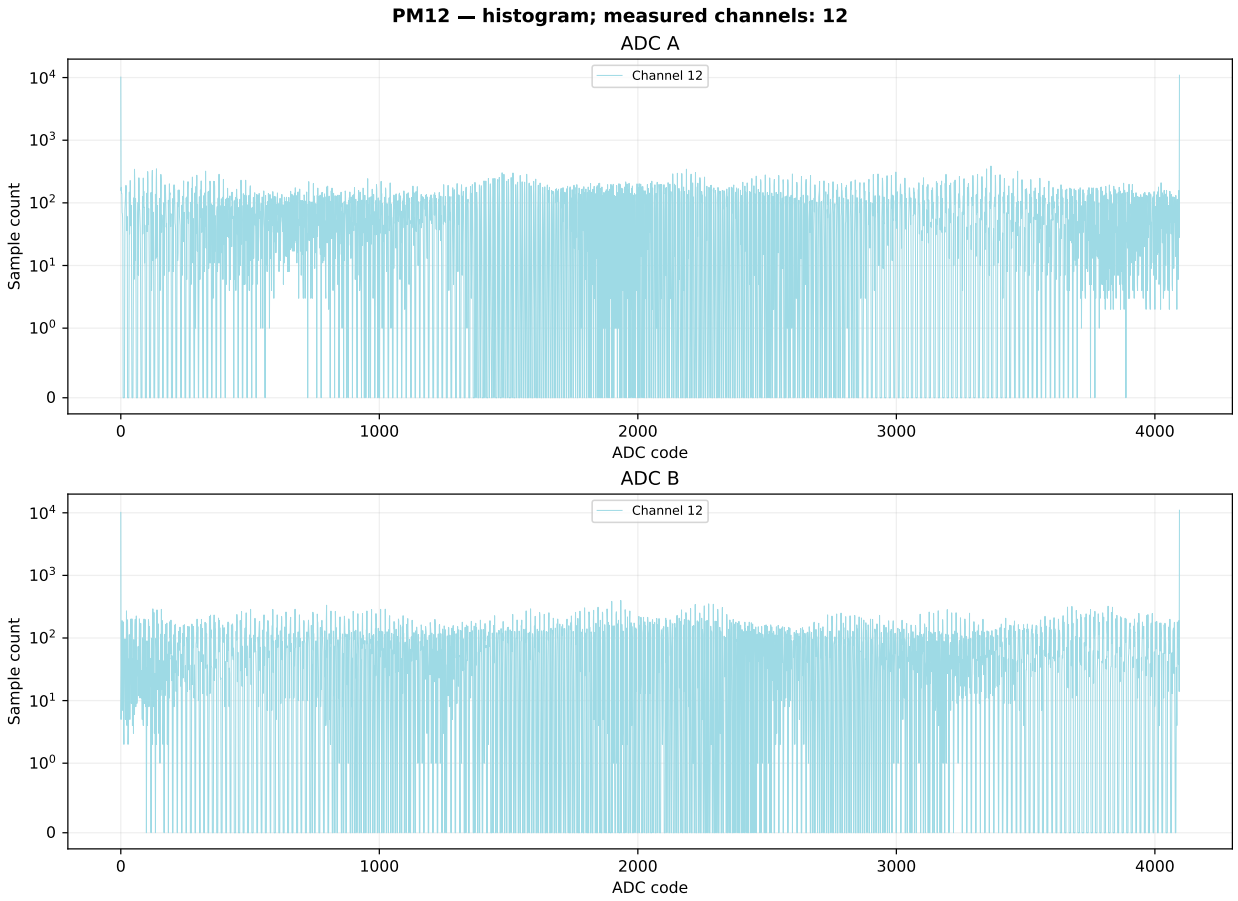
Requested DAC voltage, not a calibrated voltage measurement.



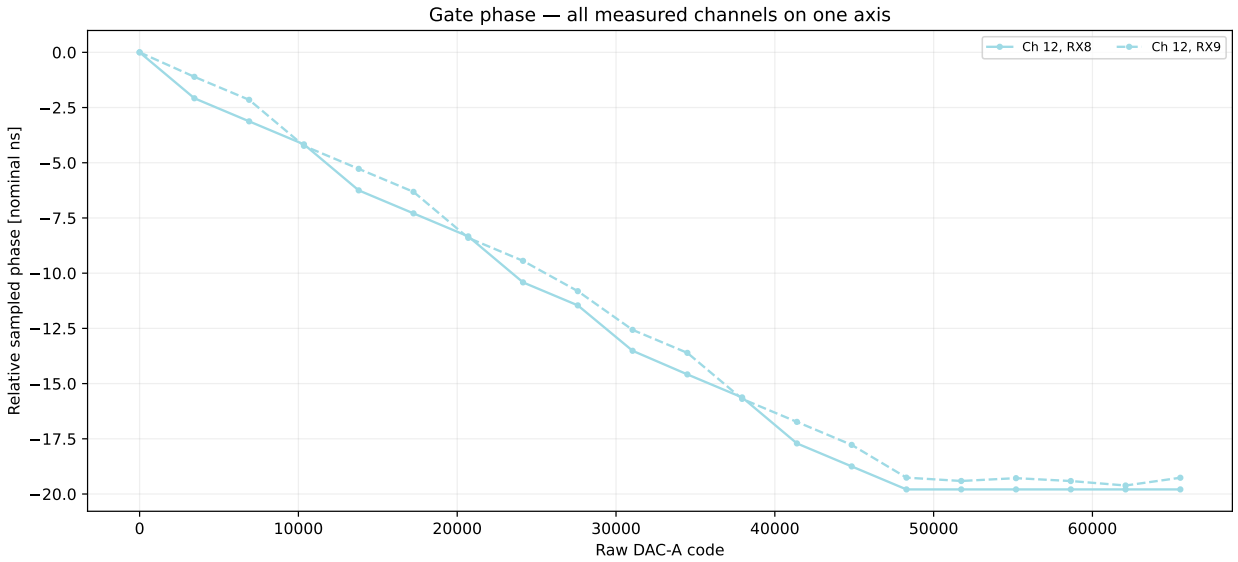
One sample per STR edge; time origins are local to sequential captures. Fs inferred as 80.1574 MHz.



DC removed, Hann window, one-sided PSD. Fs inferred from stated LHC reference, not measured by a frequency counter. No ENOB/SINAD acceptance from a DC capture.



Stepped DC code density; unvisited codes are not automatically ADC missing-code/DNL faults.



Nominal sampled-centroid units; original 20 MHz gate divider. Curves use the same reported sample metadata.

Channel	ADC	Line exercise	Visited codes	Interior samples
1	A	NOT MEASURED	-	-
1	B	NOT MEASURED	-	-
2	A	NOT MEASURED	-	-
2	B	NOT MEASURED	-	-
3	A	NOT MEASURED	-	-
3	B	NOT MEASURED	-	-
4	A	NOT MEASURED	-	-
4	B	NOT MEASURED	-	-

Channel	ADC	Line exercise	Visited codes	Interior samples
5	A	NOT MEASURED	-	-
5	B	NOT MEASURED	-	-
6	A	NOT MEASURED	-	-
6	B	NOT MEASURED	-	-
7	A	NOT MEASURED	-	-
7	B	NOT MEASURED	-	-
8	A	NOT MEASURED	-	-
8	B	NOT MEASURED	-	-
9	A	NOT MEASURED	-	-
9	B	NOT MEASURED	-	-
10	A	NOT MEASURED	-	-
10	B	NOT MEASURED	-	-
11	A	NOT MEASURED	-	-
11	B	NOT MEASURED	-	-
12	A	PASS	3129	241856
12	B	PASS	3057	241760

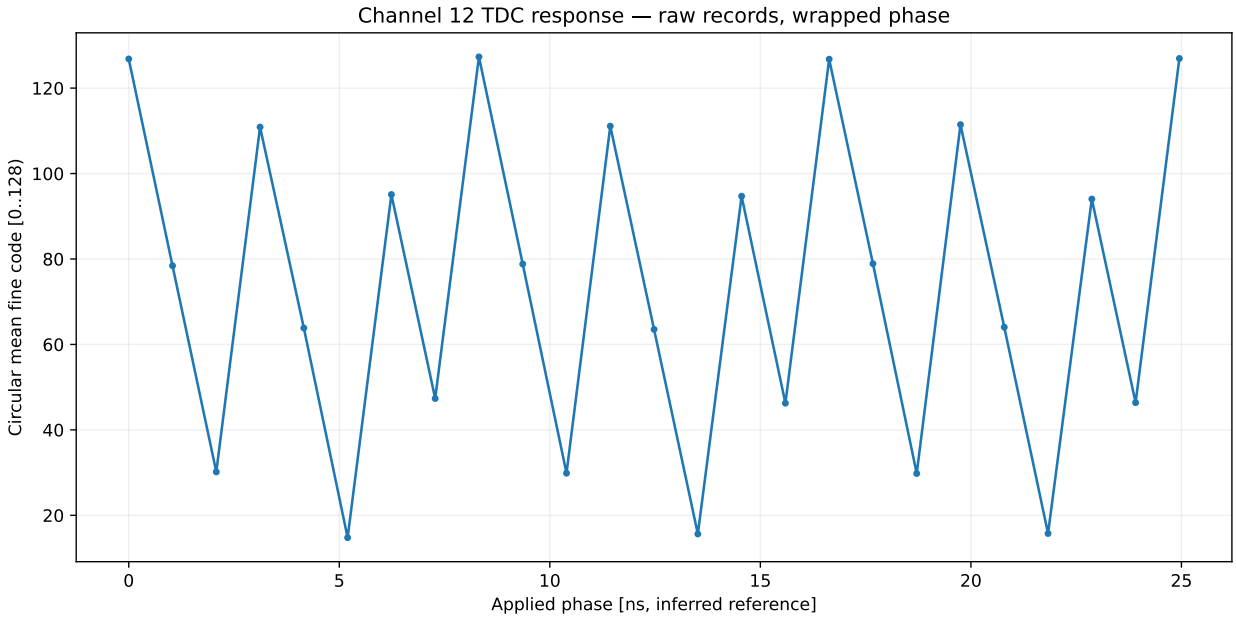
ADC	Channel	Max deviation [codes]	RMS deviation [codes]

Deviation is relative to the median of measured channels at common requested stimulus settings, without interpolation. With fewer than two channels this comparison is NOT COMPARABLE, rather than zero dispersion. Values are descriptive, not a calibrated gain/error acceptance limit.

Bit decisions exclude rail codes 0/4095. Both values of every bit and all four states of each neighboring pair must be populated: at least 64 samples and 2% of all interior samples per state, plus at least 8 samples at each of 3 distinct settings. At least 16 settings, 4096 interior samples and 3000-code mean span are required for PASS.

This excludes sampled persistent stuck/tied signatures under these conditions. It does not establish electrical resistance, bit order, timing margin, ADC INL/DNL or full missing-code compliance. A sine histogram is arcsine-shaped; a calibrated continuous ramp is a separate density test.

7 TDC phase response



Recomputed circular means from raw 13-bit TDC records. Response and event counting, not calibrated picosecond accuracy.

Applied slot	Applied delay [ns]	Fine circular mean	Events
24	0	126.828	128
25	1.03962	78.4219	128
26	2.07924	30.164	128
27	3.11886	110.883	128
28	4.15848	63.836	128
29	5.19811	14.7735	128
30	6.23773	95.0938	128
31	7.27735	47.3438	128
32	8.31697	127.312	128
33	9.35659	78.8204	128
34	10.3962	29.8593	128
35	11.4358	111.102	128
36	12.4755	63.5234	128
37	13.5151	15.6329	128
38	14.5547	94.6953	128
39	15.5943	46.2422	128
40	16.6339	126.766	128
41	17.6736	78.8984	128
42	18.7132	29.789	128
43	19.7528	111.453	128
44	20.7924	64.0389	128
45	21.832	15.7422	128
46	22.8717	94.0311	128
47	23.9113	46.375	128
48	24.9509	126.938	128

8 Internal CFD DAC B/C/D characteristics

Physical channel 12 only. First attempted restoration using computed STORED parameters was rejected; explicit active-code 0000 restoration then reproduced initial voltages. The original failed attempt remains in the source evidence.

DAC	Signal	Start [V]	End [V]
B	DRV_OFFSET	1.043846	1.003721
C	CFD_OFFSET	1.034097	1.0136713333333334
D	CFD_CMP_OFFSET	0.004514859666666667	1.0626073333333335

9 Per-link static connection test cards

Link 001/168 — STR1

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR1
FPGA package ball	W5
FPGA boundary cell	468
Expected source vector	167
Measured CPLD channel	1
Measured CPLD pin	82
Gateware discovered vector	167
Boundary discovered vector	167
Walking-one active sources	[167]
Walking-zero inactive sources	[167]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 002/168 — STR2

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR2
FPGA package ball	AA7
FPGA boundary cell	288
Expected source vector	153
Measured CPLD channel	2
Measured CPLD pin	82
Gateware discovered vector	153
Boundary discovered vector	153
Walking-one active sources	[153]
Walking-zero inactive sources	[153]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 003/168 — STR3

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR3
FPGA package ball	AA15
FPGA boundary cell	144
Expected source vector	139
Measured CPLD channel	3
Measured CPLD pin	82
Gateware discovered vector	139
Boundary discovered vector	139
Walking-one active sources	[139]
Walking-zero inactive sources	[139]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 004/168 — STR4

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR4
FPGA package ball	AD20
FPGA boundary cell	99
Expected source vector	125
Measured CPLD channel	4
Measured CPLD pin	82
Gateware discovered vector	125
Boundary discovered vector	125
Walking-one active sources	[125]
Walking-zero inactive sources	[125]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 005/168 — STR5

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR5
FPGA package ball	AB21
FPGA boundary cell	579
Expected source vector	111
Measured CPLD channel	5
Measured CPLD pin	82
Gateware discovered vector	111
Boundary discovered vector	111
Walking-one active sources	[111]
Walking-zero inactive sources	[111]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 006/168 — STR8

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR8
FPGA package ball	U20
FPGA boundary cell	726
Expected source vector	69
Measured CPLD channel	8
Measured CPLD pin	82
Gateware discovered vector	69
Boundary discovered vector	69
Walking-one active sources	[69]
Walking-zero inactive sources	[69]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 007/168 — STR9

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR9
FPGA package ball	R22
FPGA boundary cell	753
Expected source vector	55
Measured CPLD channel	9
Measured CPLD pin	82
Gateware discovered vector	55
Boundary discovered vector	55
Walking-one active sources	[55]
Walking-zero inactive sources	[55]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 008/168 — STR10

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR10
FPGA package ball	N17
FPGA boundary cell	714
Expected source vector	41
Measured CPLD channel	10
Measured CPLD pin	82
Gateware discovered vector	41
Boundary discovered vector	41
Walking-one active sources	[41]
Walking-zero inactive sources	[41]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 009/168 — STR11

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR11
FPGA package ball	M26
FPGA boundary cell	804
Expected source vector	27
Measured CPLD channel	11
Measured CPLD pin	82
Gateware discovered vector	27
Boundary discovered vector	27
Walking-one active sources	[27]
Walking-zero inactive sources	[27]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 010/168 — STR12

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR12
FPGA package ball	J19
FPGA boundary cell	1014
Expected source vector	13
Measured CPLD channel	12
Measured CPLD pin	82
Gateware discovered vector	13
Boundary discovered vector	13
Walking-one active sources	[13]
Walking-zero inactive sources	[13]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 011/168 — STR6

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR6
FPGA package ball	AA23
FPGA boundary cell	621
Expected source vector	97
Measured CPLD channel	6
Measured CPLD pin	82
Gateware discovered vector	97
Boundary discovered vector	97
Walking-one active sources	[97]
Walking-zero inactive sources	[97]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 012/168 — STR7

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	STR7
FPGA package ball	W26
FPGA boundary cell	654
Expected source vector	83
Measured CPLD channel	7
Measured CPLD pin	82
Gateware discovered vector	83
Boundary discovered vector	83
Walking-one active sources	[83]
Walking-zero inactive sources	[83]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 013/168 — DI1[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[8]
FPGA package ball	V4
FPGA boundary cell	453
Expected source vector	162
Measured CPLD channel	1
Measured CPLD pin	90
Gateware discovered vector	162
Boundary discovered vector	162
Walking-one active sources	[162]
Walking-zero inactive sources	[162]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 014/168 — DI1[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[9]
FPGA package ball	W4
FPGA boundary cell	450
Expected source vector	163
Measured CPLD channel	1
Measured CPLD pin	89
Gateware discovered vector	163
Boundary discovered vector	163
Walking-one active sources	[163]
Walking-zero inactive sources	[163]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 015/168 — DI1[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[12]
FPGA package ball	W6
FPGA boundary cell	471
Expected source vector	166
Measured CPLD channel	1
Measured CPLD pin	85
Gateware discovered vector	166
Boundary discovered vector	166
Walking-one active sources	[166]
Walking-zero inactive sources	[166]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 016/168 — DI2[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[4]
FPGA package ball	AB6
FPGA boundary cell	393
Expected source vector	144
Measured CPLD channel	2
Measured CPLD pin	94
Gateware discovered vector	144
Boundary discovered vector	144
Walking-one active sources	[144]
Walking-zero inactive sources	[144]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 017/168 — DI2[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[8]
FPGA package ball	U9
FPGA boundary cell	336
Expected source vector	148
Measured CPLD channel	2
Measured CPLD pin	90
Gateware discovered vector	148
Boundary discovered vector	148
Walking-one active sources	[148]
Walking-zero inactive sources	[148]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 018/168 — DI2[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[9]
FPGA package ball	V7
FPGA boundary cell	324
Expected source vector	149
Measured CPLD channel	2
Measured CPLD pin	89
Gateware discovered vector	149
Boundary discovered vector	149
Walking-one active sources	[149]
Walking-zero inactive sources	[149]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 019/168 — DI2[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[10]
FPGA package ball	V8
FPGA boundary cell	327
Expected source vector	150
Measured CPLD channel	2
Measured CPLD pin	87
Gateware discovered vector	150
Boundary discovered vector	150
Walking-one active sources	[150]
Walking-zero inactive sources	[150]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 020/168 — DI2[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[11]
FPGA package ball	Y7
FPGA boundary cell	312
Expected source vector	151
Measured CPLD channel	2
Measured CPLD pin	86
Gateware discovered vector	151
Boundary discovered vector	151
Walking-one active sources	[151]
Walking-zero inactive sources	[151]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 021/168 — DI2[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[12]
FPGA package ball	Y8
FPGA boundary cell	315
Expected source vector	152
Measured CPLD channel	2
Measured CPLD pin	85
Gateware discovered vector	152
Boundary discovered vector	152
Walking-one active sources	[152]
Walking-zero inactive sources	[152]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 022/168 — DI3[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[8]
FPGA package ball	V13
FPGA boundary cell	186
Expected source vector	134
Measured CPLD channel	3
Measured CPLD pin	90
Gateware discovered vector	134
Boundary discovered vector	134
Walking-one active sources	[134]
Walking-zero inactive sources	[134]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 023/168 — DI3[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[9]
FPGA package ball	W13
FPGA boundary cell	39
Expected source vector	135
Measured CPLD channel	3
Measured CPLD pin	89
Gateware discovered vector	135
Boundary discovered vector	135
Walking-one active sources	[135]
Walking-zero inactive sources	[135]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 024/168 — DI3[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[10]
FPGA package ball	V14
FPGA boundary cell	45
Expected source vector	136
Measured CPLD channel	3
Measured CPLD pin	87
Gateware discovered vector	136
Boundary discovered vector	136
Walking-one active sources	[136]
Walking-zero inactive sources	[136]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 025/168 — DI3[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[11]
FPGA package ball	W14
FPGA boundary cell	42
Expected source vector	137
Measured CPLD channel	3
Measured CPLD pin	86
Gateware discovered vector	137
Boundary discovered vector	137
Walking-one active sources	[137]
Walking-zero inactive sources	[137]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 026/168 — DI3[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[12]
FPGA package ball	AA14
FPGA boundary cell	147
Expected source vector	138
Measured CPLD channel	3
Measured CPLD pin	85
Gateware discovered vector	138
Boundary discovered vector	138
Walking-one active sources	[138]
Walking-zero inactive sources	[138]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 027/168 — DI4[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[0]
FPGA package ball	V18
FPGA boundary cell	51
Expected source vector	112
Measured CPLD channel	4
Measured CPLD pin	99
Gateware discovered vector	112
Boundary discovered vector	112
Walking-one active sources	[112]
Walking-zero inactive sources	[112]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 028/168 — DI4[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[1]
FPGA package ball	W18
FPGA boundary cell	63
Expected source vector	113
Measured CPLD channel	4
Measured CPLD pin	97
Gateware discovered vector	113
Boundary discovered vector	113
Walking-one active sources	[113]
Walking-zero inactive sources	[113]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 029/168 — DI4[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[4]
FPGA package ball	AE18
FPGA boundary cell	171
Expected source vector	116
Measured CPLD channel	4
Measured CPLD pin	94
Gateware discovered vector	116
Boundary discovered vector	116
Walking-one active sources	[116]
Walking-zero inactive sources	[116]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 030/168 — DI4[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[5]
FPGA package ball	AF18
FPGA boundary cell	168
Expected source vector	117
Measured CPLD channel	4
Measured CPLD pin	93
Gateware discovered vector	117
Boundary discovered vector	117
Walking-one active sources	[117]
Walking-zero inactive sources	[117]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 031/168 — DI4[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[7]
FPGA package ball	V19
FPGA boundary cell	48
Expected source vector	119
Measured CPLD channel	4
Measured CPLD pin	91
Gateware discovered vector	119
Boundary discovered vector	119
Walking-one active sources	[119]
Walking-zero inactive sources	[119]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 032/168 — DI4[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[9]
FPGA package ball	AB19
FPGA boundary cell	81
Expected source vector	121
Measured CPLD channel	4
Measured CPLD pin	89
Gateware discovered vector	121
Boundary discovered vector	121
Walking-one active sources	[121]
Walking-zero inactive sources	[121]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 033/168 — DI4[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[10]
FPGA package ball	AD19
FPGA boundary cell	84
Expected source vector	122
Measured CPLD channel	4
Measured CPLD pin	87
Gateware discovered vector	122
Boundary discovered vector	122
Walking-one active sources	[122]
Walking-zero inactive sources	[122]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 034/168 — DI4[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[11]
FPGA package ball	AF19
FPGA boundary cell	159
Expected source vector	123
Measured CPLD channel	4
Measured CPLD pin	86
Gateware discovered vector	123
Boundary discovered vector	123
Walking-one active sources	[123]
Walking-zero inactive sources	[123]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 035/168 — DI5[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[0]
FPGA package ball	AE20
FPGA boundary cell	96
Expected source vector	98
Measured CPLD channel	5
Measured CPLD pin	99
Gateware discovered vector	98
Boundary discovered vector	98
Walking-one active sources	[98]
Walking-zero inactive sources	[98]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 036/168 — DI5[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[1]
FPGA package ball	AF20
FPGA boundary cell	156
Expected source vector	99
Measured CPLD channel	5
Measured CPLD pin	97
Gateware discovered vector	99
Boundary discovered vector	99
Walking-one active sources	[99]
Walking-zero inactive sources	[99]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 037/168 — DI5[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[2]
FPGA package ball	AE21
FPGA boundary cell	570
Expected source vector	100
Measured CPLD channel	5
Measured CPLD pin	96
Gateware discovered vector	100
Boundary discovered vector	100
Walking-one active sources	[100]
Walking-zero inactive sources	[100]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 038/168 — DI5[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[3]
FPGA package ball	AF22
FPGA boundary cell	540
Expected source vector	101
Measured CPLD channel	5
Measured CPLD pin	95
Gateware discovered vector	101
Boundary discovered vector	101
Walking-one active sources	[101]
Walking-zero inactive sources	[101]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 039/168 — DI5[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[4]
FPGA package ball	AE22
FPGA boundary cell	543
Expected source vector	102
Measured CPLD channel	5
Measured CPLD pin	94
Gateware discovered vector	102
Boundary discovered vector	102
Walking-one active sources	[102]
Walking-zero inactive sources	[102]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 040/168 — DI5[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[5]
FPGA package ball	AF23
FPGA boundary cell	552
Expected source vector	103
Measured CPLD channel	5
Measured CPLD pin	93
Gateware discovered vector	103
Boundary discovered vector	103
Walking-one active sources	[103]
Walking-zero inactive sources	[103]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 041/168 — DI5[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[6]
FPGA package ball	AE23
FPGA boundary cell	555
Expected source vector	104
Measured CPLD channel	5
Measured CPLD pin	92
Gateware discovered vector	104
Boundary discovered vector	104
Walking-one active sources	[104]
Walking-zero inactive sources	[104]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 042/168 — DI5[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[7]
FPGA package ball	AF24
FPGA boundary cell	567
Expected source vector	105
Measured CPLD channel	5
Measured CPLD pin	91
Gateware discovered vector	105
Boundary discovered vector	105
Walking-one active sources	[105]
Walking-zero inactive sources	[105]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 043/168 — DI5[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[8]
FPGA package ball	AF25
FPGA boundary cell	564
Expected source vector	106
Measured CPLD channel	5
Measured CPLD pin	90
Gateware discovered vector	106
Boundary discovered vector	106
Walking-one active sources	[106]
Walking-zero inactive sources	[106]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 044/168 — DI5[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[9]
FPGA package ball	AE25
FPGA boundary cell	546
Expected source vector	107
Measured CPLD channel	5
Measured CPLD pin	89
Gateware discovered vector	107
Boundary discovered vector	107
Walking-one active sources	[107]
Walking-zero inactive sources	[107]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 045/168 — DI5[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[12]
FPGA package ball	AD23
FPGA boundary cell	591
Expected source vector	110
Measured CPLD channel	5
Measured CPLD pin	85
Gateware discovered vector	110
Boundary discovered vector	110
Walking-one active sources	[110]
Walking-zero inactive sources	[110]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 046/168 — DI6[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[6]
FPGA package ball	AC26
FPGA boundary cell	630
Expected source vector	90
Measured CPLD channel	6
Measured CPLD pin	92
Gateware discovered vector	90
Boundary discovered vector	90
Walking-one active sources	[90]
Walking-zero inactive sources	[90]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 047/168 — DI6[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[8]
FPGA package ball	AB22
FPGA boundary cell	585
Expected source vector	92
Measured CPLD channel	6
Measured CPLD pin	90
Gateware discovered vector	92
Boundary discovered vector	92
Walking-one active sources	[92]
Walking-zero inactive sources	[92]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 048/168 — DI7[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[0]
FPGA package ball	AA24
FPGA boundary cell	612
Expected source vector	70
Measured CPLD channel	7
Measured CPLD pin	99
Gateware discovered vector	70
Boundary discovered vector	70
Walking-one active sources	[70]
Walking-zero inactive sources	[70]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 049/168 — DI7[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[1]
FPGA package ball	AA25
FPGA boundary cell	645
Expected source vector	71
Measured CPLD channel	7
Measured CPLD pin	97
Gateware discovered vector	71
Boundary discovered vector	71
Walking-one active sources	[71]
Walking-zero inactive sources	[71]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 050/168 — DI7[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[2]
FPGA package ball	Y20
FPGA boundary cell	537
Expected source vector	72
Measured CPLD channel	7
Measured CPLD pin	96
Gateware discovered vector	72
Boundary discovered vector	72
Walking-one active sources	[72]
Walking-zero inactive sources	[72]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 051/168 — DI7[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[8]
FPGA package ball	W20
FPGA boundary cell	597
Expected source vector	78
Measured CPLD channel	7
Measured CPLD pin	90
Gateware discovered vector	78
Boundary discovered vector	78
Walking-one active sources	[78]
Walking-zero inactive sources	[78]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 052/168 — DI7[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[9]
FPGA package ball	W21
FPGA boundary cell	648
Expected source vector	79
Measured CPLD channel	7
Measured CPLD pin	89
Gateware discovered vector	79
Boundary discovered vector	79
Walking-one active sources	[79]
Walking-zero inactive sources	[79]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 053/168 — DI8[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[9]
FPGA package ball	U26
FPGA boundary cell	663
Expected source vector	65
Measured CPLD channel	8
Measured CPLD pin	89
Gateware discovered vector	65
Boundary discovered vector	65
Walking-one active sources	[65]
Walking-zero inactive sources	[65]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 054/168 — DI8[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[10]
FPGA package ball	U16
FPGA boundary cell	687
Expected source vector	66
Measured CPLD channel	8
Measured CPLD pin	87
Gateware discovered vector	66
Boundary discovered vector	66
Walking-one active sources	[66]
Walking-zero inactive sources	[66]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 055/168 — DI8[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[11]
FPGA package ball	U17
FPGA boundary cell	699
Expected source vector	67
Measured CPLD channel	8
Measured CPLD pin	86
Gateware discovered vector	67
Boundary discovered vector	67
Walking-one active sources	[67]
Walking-zero inactive sources	[67]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 056/168 — DI8[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[12]
FPGA package ball	U19
FPGA boundary cell	729
Expected source vector	68
Measured CPLD channel	8
Measured CPLD pin	85
Gateware discovered vector	68
Boundary discovered vector	68
Walking-one active sources	[68]
Walking-zero inactive sources	[68]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 057/168 — DI9[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[0]
FPGA package ball	T17
FPGA boundary cell	696
Expected source vector	42
Measured CPLD channel	9
Measured CPLD pin	99
Gateware discovered vector	42
Boundary discovered vector	42
Walking-one active sources	[42]
Walking-zero inactive sources	[42]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 058/168 — DI9[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[1]
FPGA package ball	T18
FPGA boundary cell	723
Expected source vector	43
Measured CPLD channel	9
Measured CPLD pin	97
Gateware discovered vector	43
Boundary discovered vector	43
Walking-one active sources	[43]
Walking-zero inactive sources	[43]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 059/168 — DI9[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[2]
FPGA package ball	T19
FPGA boundary cell	720
Expected source vector	44
Measured CPLD channel	9
Measured CPLD pin	96
Gateware discovered vector	44
Boundary discovered vector	44
Walking-one active sources	[44]
Walking-zero inactive sources	[44]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 060/168 — DI9[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[3]
FPGA package ball	T20
FPGA boundary cell	741
Expected source vector	45
Measured CPLD channel	9
Measured CPLD pin	95
Gateware discovered vector	45
Boundary discovered vector	45
Walking-one active sources	[45]
Walking-zero inactive sources	[45]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 061/168 — DI9[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[4]
FPGA package ball	T22
FPGA boundary cell	735
Expected source vector	46
Measured CPLD channel	9
Measured CPLD pin	94
Gateware discovered vector	46
Boundary discovered vector	46
Walking-one active sources	[46]
Walking-zero inactive sources	[46]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 062/168 — DI9[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[5]
FPGA package ball	T23
FPGA boundary cell	732
Expected source vector	47
Measured CPLD channel	9
Measured CPLD pin	93
Gateware discovered vector	47
Boundary discovered vector	47
Walking-one active sources	[47]
Walking-zero inactive sources	[47]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 063/168 — DI9[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[6]
FPGA package ball	T24
FPGA boundary cell	747
Expected source vector	48
Measured CPLD channel	9
Measured CPLD pin	92
Gateware discovered vector	48
Boundary discovered vector	48
Walking-one active sources	[48]
Walking-zero inactive sources	[48]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 064/168 — DI9[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[7]
FPGA package ball	T25
FPGA boundary cell	744
Expected source vector	49
Measured CPLD channel	9
Measured CPLD pin	91
Gateware discovered vector	49
Boundary discovered vector	49
Walking-one active sources	[49]
Walking-zero inactive sources	[49]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 065/168 — DI9[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[8]
FPGA package ball	R16
FPGA boundary cell	711
Expected source vector	50
Measured CPLD channel	9
Measured CPLD pin	90
Gateware discovered vector	50
Boundary discovered vector	50
Walking-one active sources	[50]
Walking-zero inactive sources	[50]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 066/168 — DI9[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[9]
FPGA package ball	R17
FPGA boundary cell	708
Expected source vector	51
Measured CPLD channel	9
Measured CPLD pin	89
Gateware discovered vector	51
Boundary discovered vector	51
Walking-one active sources	[51]
Walking-zero inactive sources	[51]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 067/168 — DI9[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[10]
FPGA package ball	R18
FPGA boundary cell	693
Expected source vector	52
Measured CPLD channel	9
Measured CPLD pin	87
Gateware discovered vector	52
Boundary discovered vector	52
Walking-one active sources	[52]
Walking-zero inactive sources	[52]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 068/168 — DI9[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[11]
FPGA package ball	R20
FPGA boundary cell	738
Expected source vector	53
Measured CPLD channel	9
Measured CPLD pin	86
Gateware discovered vector	53
Boundary discovered vector	53
Walking-one active sources	[53]
Walking-zero inactive sources	[53]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 069/168 — DI9[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI9[12]
FPGA package ball	R21
FPGA boundary cell	759
Expected source vector	54
Measured CPLD channel	9
Measured CPLD pin	85
Gateware discovered vector	54
Boundary discovered vector	54
Walking-one active sources	[54]
Walking-zero inactive sources	[54]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 070/168 — DI10[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[0]
FPGA package ball	R23
FPGA boundary cell	750
Expected source vector	28
Measured CPLD channel	10
Measured CPLD pin	99
Gateware discovered vector	28
Boundary discovered vector	28
Walking-one active sources	[28]
Walking-zero inactive sources	[28]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 071/168 — DI10[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[1]
FPGA package ball	R25
FPGA boundary cell	801
Expected source vector	29
Measured CPLD channel	10
Measured CPLD pin	97
Gateware discovered vector	29
Boundary discovered vector	29
Walking-one active sources	[29]
Walking-zero inactive sources	[29]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 072/168 — DI10[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[2]
FPGA package ball	R26
FPGA boundary cell	825
Expected source vector	30
Measured CPLD channel	10
Measured CPLD pin	96
Gateware discovered vector	30
Boundary discovered vector	30
Walking-one active sources	[30]
Walking-zero inactive sources	[30]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 073/168 — DI10[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[3]
FPGA package ball	P16
FPGA boundary cell	717
Expected source vector	31
Measured CPLD channel	10
Measured CPLD pin	95
Gateware discovered vector	31
Boundary discovered vector	31
Walking-one active sources	[31]
Walking-zero inactive sources	[31]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 074/168 — DI10[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[4]
FPGA package ball	P18
FPGA boundary cell	690
Expected source vector	32
Measured CPLD channel	10
Measured CPLD pin	94
Gateware discovered vector	32
Boundary discovered vector	32
Walking-one active sources	[32]
Walking-zero inactive sources	[32]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 075/168 — DI10[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[5]
FPGA package ball	P19
FPGA boundary cell	783
Expected source vector	33
Measured CPLD channel	10
Measured CPLD pin	93
Gateware discovered vector	33
Boundary discovered vector	33
Walking-one active sources	[33]
Walking-zero inactive sources	[33]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 076/168 — DI10[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[6]
FPGA package ball	P20
FPGA boundary cell	780
Expected source vector	34
Measured CPLD channel	10
Measured CPLD pin	92
Gateware discovered vector	34
Boundary discovered vector	34
Walking-one active sources	[34]
Walking-zero inactive sources	[34]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 077/168 — DI10[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[7]
FPGA package ball	P21
FPGA boundary cell	756
Expected source vector	35
Measured CPLD channel	10
Measured CPLD pin	91
Gateware discovered vector	35
Boundary discovered vector	35
Walking-one active sources	[35]
Walking-zero inactive sources	[35]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 078/168 — DI10[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[8]
FPGA package ball	P23
FPGA boundary cell	771
Expected source vector	36
Measured CPLD channel	10
Measured CPLD pin	90
Gateware discovered vector	36
Boundary discovered vector	36
Walking-one active sources	[36]
Walking-zero inactive sources	[36]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 079/168 — DI10[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[9]
FPGA package ball	P24
FPGA boundary cell	813
Expected source vector	37
Measured CPLD channel	10
Measured CPLD pin	89
Gateware discovered vector	37
Boundary discovered vector	37
Walking-one active sources	[37]
Walking-zero inactive sources	[37]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 080/168 — DI10[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[10]
FPGA package ball	P25
FPGA boundary cell	798
Expected source vector	38
Measured CPLD channel	10
Measured CPLD pin	87
Gateware discovered vector	38
Boundary discovered vector	38
Walking-one active sources	[38]
Walking-zero inactive sources	[38]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 081/168 — DI10[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[11]
FPGA package ball	P26
FPGA boundary cell	822
Expected source vector	39
Measured CPLD channel	10
Measured CPLD pin	86
Gateware discovered vector	39
Boundary discovered vector	39
Walking-one active sources	[39]
Walking-zero inactive sources	[39]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 082/168 — DI10[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI10[12]
FPGA package ball	N16
FPGA boundary cell	834
Expected source vector	40
Measured CPLD channel	10
Measured CPLD pin	85
Gateware discovered vector	40
Boundary discovered vector	40
Walking-one active sources	[40]
Walking-zero inactive sources	[40]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 083/168 — DI11[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[0]
FPGA package ball	N18
FPGA boundary cell	705
Expected source vector	14
Measured CPLD channel	11
Measured CPLD pin	99
Gateware discovered vector	14
Boundary discovered vector	14
Walking-one active sources	[14]
Walking-zero inactive sources	[14]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 084/168 — DI11[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[1]
FPGA package ball	N19
FPGA boundary cell	795
Expected source vector	15
Measured CPLD channel	11
Measured CPLD pin	97
Gateware discovered vector	15
Boundary discovered vector	15
Walking-one active sources	[15]
Walking-zero inactive sources	[15]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 085/168 — DI11[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[2]
FPGA package ball	N21
FPGA boundary cell	765
Expected source vector	16
Measured CPLD channel	11
Measured CPLD pin	96
Gateware discovered vector	16
Boundary discovered vector	16
Walking-one active sources	[16]
Walking-zero inactive sources	[16]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 086/168 — DI11[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[3]
FPGA package ball	N22
FPGA boundary cell	762
Expected source vector	17
Measured CPLD channel	11
Measured CPLD pin	95
Gateware discovered vector	17
Boundary discovered vector	17
Walking-one active sources	[17]
Walking-zero inactive sources	[17]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 087/168 — DI11[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[4]
FPGA package ball	N23
FPGA boundary cell	768
Expected source vector	18
Measured CPLD channel	11
Measured CPLD pin	94
Gateware discovered vector	18
Boundary discovered vector	18
Walking-one active sources	[18]
Walking-zero inactive sources	[18]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 088/168 — DI11[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[5]
FPGA package ball	N24
FPGA boundary cell	810
Expected source vector	19
Measured CPLD channel	11
Measured CPLD pin	93
Gateware discovered vector	19
Boundary discovered vector	19
Walking-one active sources	[19]
Walking-zero inactive sources	[19]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 089/168 — DI11[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[6]
FPGA package ball	N26
FPGA boundary cell	807
Expected source vector	20
Measured CPLD channel	11
Measured CPLD pin	92
Gateware discovered vector	20
Boundary discovered vector	20
Walking-one active sources	[20]
Walking-zero inactive sources	[20]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 090/168 — DI11[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[7]
FPGA package ball	M19
FPGA boundary cell	702
Expected source vector	21
Measured CPLD channel	11
Measured CPLD pin	91
Gateware discovered vector	21
Boundary discovered vector	21
Walking-one active sources	[21]
Walking-zero inactive sources	[21]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 091/168 — DI11[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[8]
FPGA package ball	M20
FPGA boundary cell	792
Expected source vector	22
Measured CPLD channel	11
Measured CPLD pin	90
Gateware discovered vector	22
Boundary discovered vector	22
Walking-one active sources	[22]
Walking-zero inactive sources	[22]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 092/168 — DI11[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[9]
FPGA package ball	M21
FPGA boundary cell	777
Expected source vector	23
Measured CPLD channel	11
Measured CPLD pin	89
Gateware discovered vector	23
Boundary discovered vector	23
Walking-one active sources	[23]
Walking-zero inactive sources	[23]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 093/168 — DI11[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[10]
FPGA package ball	M22
FPGA boundary cell	774
Expected source vector	24
Measured CPLD channel	11
Measured CPLD pin	87
Gateware discovered vector	24
Boundary discovered vector	24
Walking-one active sources	[24]
Walking-zero inactive sources	[24]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 094/168 — DI11[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[11]
FPGA package ball	M24
FPGA boundary cell	789
Expected source vector	25
Measured CPLD channel	11
Measured CPLD pin	86
Gateware discovered vector	25
Boundary discovered vector	25
Walking-one active sources	[25]
Walking-zero inactive sources	[25]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 095/168 — DI11[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI11[12]
FPGA package ball	M25
FPGA boundary cell	819
Expected source vector	26
Measured CPLD channel	11
Measured CPLD pin	85
Gateware discovered vector	26
Boundary discovered vector	26
Walking-one active sources	[26]
Walking-zero inactive sources	[26]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 096/168 — DI12[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[0]
FPGA package ball	L24
FPGA boundary cell	786
Expected source vector	0
Measured CPLD channel	12
Measured CPLD pin	99
Gateware discovered vector	0
Boundary discovered vector	0
Walking-one active sources	[0]
Walking-zero inactive sources	[0]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 097/168 — DI12[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[1]
FPGA package ball	L25
FPGA boundary cell	816
Expected source vector	1
Measured CPLD channel	12
Measured CPLD pin	97
Gateware discovered vector	1
Boundary discovered vector	1
Walking-one active sources	[1]
Walking-zero inactive sources	[1]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 098/168 — DI12[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[2]
FPGA package ball	K25
FPGA boundary cell	831
Expected source vector	2
Measured CPLD channel	12
Measured CPLD pin	96
Gateware discovered vector	2
Boundary discovered vector	2
Walking-one active sources	[2]
Walking-zero inactive sources	[2]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 099/168 — DI12[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[3]
FPGA package ball	K26
FPGA boundary cell	828
Expected source vector	3
Measured CPLD channel	12
Measured CPLD pin	95
Gateware discovered vector	3
Boundary discovered vector	3
Walking-one active sources	[3]
Walking-zero inactive sources	[3]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 100/168 — DI12[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[4]
FPGA package ball	M16
FPGA boundary cell	987
Expected source vector	4
Measured CPLD channel	12
Measured CPLD pin	94
Gateware discovered vector	4
Boundary discovered vector	4
Walking-one active sources	[4]
Walking-zero inactive sources	[4]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 101/168 — DI12[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[5]
FPGA package ball	M17
FPGA boundary cell	999
Expected source vector	5
Measured CPLD channel	12
Measured CPLD pin	93
Gateware discovered vector	5
Boundary discovered vector	5
Walking-one active sources	[5]
Walking-zero inactive sources	[5]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 102/168 — DI12[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[6]
FPGA package ball	L17
FPGA boundary cell	993
Expected source vector	6
Measured CPLD channel	12
Measured CPLD pin	92
Gateware discovered vector	6
Boundary discovered vector	6
Walking-one active sources	[6]
Walking-zero inactive sources	[6]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 103/168 — DI12[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[7]
FPGA package ball	L18
FPGA boundary cell	996
Expected source vector	7
Measured CPLD channel	12
Measured CPLD pin	91
Gateware discovered vector	7
Boundary discovered vector	7
Walking-one active sources	[7]
Walking-zero inactive sources	[7]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 104/168 — DI12[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[8]
FPGA package ball	L19
FPGA boundary cell	1011
Expected source vector	8
Measured CPLD channel	12
Measured CPLD pin	90
Gateware discovered vector	8
Boundary discovered vector	8
Walking-one active sources	[8]
Walking-zero inactive sources	[8]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 105/168 — DI12[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[9]
FPGA package ball	L20
FPGA boundary cell	1008
Expected source vector	9
Measured CPLD channel	12
Measured CPLD pin	89
Gateware discovered vector	9
Boundary discovered vector	9
Walking-one active sources	[9]
Walking-zero inactive sources	[9]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 106/168 — DI12[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[10]
FPGA package ball	K18
FPGA boundary cell	990
Expected source vector	10
Measured CPLD channel	12
Measured CPLD pin	87
Gateware discovered vector	10
Boundary discovered vector	10
Walking-one active sources	[10]
Walking-zero inactive sources	[10]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 107/168 — DI12[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[11]
FPGA package ball	K16
FPGA boundary cell	1005
Expected source vector	11
Measured CPLD channel	12
Measured CPLD pin	86
Gateware discovered vector	11
Boundary discovered vector	11
Walking-one active sources	[11]
Walking-zero inactive sources	[11]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 108/168 — DI12[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI12[12]
FPGA package ball	K17
FPGA boundary cell	1002
Expected source vector	12
Measured CPLD channel	12
Measured CPLD pin	85
Gateware discovered vector	12
Boundary discovered vector	12
Walking-one active sources	[12]
Walking-zero inactive sources	[12]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 109/168 — DI1[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[0]
FPGA package ball	T7
FPGA boundary cell	339
Expected source vector	154
Measured CPLD channel	1
Measured CPLD pin	99
Gateware discovered vector	154
Boundary discovered vector	154
Walking-one active sources	[154]
Walking-zero inactive sources	[154]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 110/168 — DI1[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[1]
FPGA package ball	U7
FPGA boundary cell	459
Expected source vector	155
Measured CPLD channel	1
Measured CPLD pin	97
Gateware discovered vector	155
Boundary discovered vector	155
Walking-one active sources	[155]
Walking-zero inactive sources	[155]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 111/168 — DI1[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[2]
FPGA package ball	U6
FPGA boundary cell	483
Expected source vector	156
Measured CPLD channel	1
Measured CPLD pin	96
Gateware discovered vector	156
Boundary discovered vector	156
Walking-one active sources	[156]
Walking-zero inactive sources	[156]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 112/168 — DI1[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[3]
FPGA package ball	U5
FPGA boundary cell	480
Expected source vector	157
Measured CPLD channel	1
Measured CPLD pin	95
Gateware discovered vector	157
Boundary discovered vector	157
Walking-one active sources	[157]
Walking-zero inactive sources	[157]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 113/168 — DI1[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[4]
FPGA package ball	U4
FPGA boundary cell	486
Expected source vector	158
Measured CPLD channel	1
Measured CPLD pin	94
Gateware discovered vector	158
Boundary discovered vector	158
Walking-one active sources	[158]
Walking-zero inactive sources	[158]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 114/168 — DI1[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[5]
FPGA package ball	U2
FPGA boundary cell	477
Expected source vector	159
Measured CPLD channel	1
Measured CPLD pin	93
Gateware discovered vector	159
Boundary discovered vector	159
Walking-one active sources	[159]
Walking-zero inactive sources	[159]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 115/168 — DI1[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[6]
FPGA package ball	U1
FPGA boundary cell	474
Expected source vector	160
Measured CPLD channel	1
Measured CPLD pin	92
Gateware discovered vector	160
Boundary discovered vector	160
Walking-one active sources	[160]
Walking-zero inactive sources	[160]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 116/168 — DI1[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[7]
FPGA package ball	V6
FPGA boundary cell	456
Expected source vector	161
Measured CPLD channel	1
Measured CPLD pin	91
Gateware discovered vector	161
Boundary discovered vector	161
Walking-one active sources	[161]
Walking-zero inactive sources	[161]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 117/168 — DI1[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[10]
FPGA package ball	V2
FPGA boundary cell	441
Expected source vector	164
Measured CPLD channel	1
Measured CPLD pin	87
Gateware discovered vector	164
Boundary discovered vector	164
Walking-one active sources	[164]
Walking-zero inactive sources	[164]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 118/168 — DI1[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI1[11]
FPGA package ball	V1
FPGA boundary cell	438
Expected source vector	165
Measured CPLD channel	1
Measured CPLD pin	86
Gateware discovered vector	165
Boundary discovered vector	165
Walking-one active sources	[165]
Walking-zero inactive sources	[165]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 119/168 — DI2[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[0]
FPGA package ball	AF4
FPGA boundary cell	360
Expected source vector	140
Measured CPLD channel	2
Measured CPLD pin	99
Gateware discovered vector	140
Boundary discovered vector	140
Walking-one active sources	[140]
Walking-zero inactive sources	[140]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 120/168 — DI2[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[1]
FPGA package ball	AD5
FPGA boundary cell	378
Expected source vector	141
Measured CPLD channel	2
Measured CPLD pin	97
Gateware discovered vector	141
Boundary discovered vector	141
Walking-one active sources	[141]
Walking-zero inactive sources	[141]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 121/168 — DI2[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[2]
FPGA package ball	AE5
FPGA boundary cell	348
Expected source vector	142
Measured CPLD channel	2
Measured CPLD pin	96
Gateware discovered vector	142
Boundary discovered vector	142
Walking-one active sources	[142]
Walking-zero inactive sources	[142]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 122/168 — DI2[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[3]
FPGA package ball	AF5
FPGA boundary cell	363
Expected source vector	143
Measured CPLD channel	2
Measured CPLD pin	95
Gateware discovered vector	143
Boundary discovered vector	143
Walking-one active sources	[143]
Walking-zero inactive sources	[143]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 123/168 — DI2[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[5]
FPGA package ball	AC6
FPGA boundary cell	390
Expected source vector	145
Measured CPLD channel	2
Measured CPLD pin	93
Gateware discovered vector	145
Boundary discovered vector	145
Walking-one active sources	[145]
Walking-zero inactive sources	[145]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 124/168 — DI2[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[6]
FPGA package ball	AD6
FPGA boundary cell	381
Expected source vector	146
Measured CPLD channel	2
Measured CPLD pin	92
Gateware discovered vector	146
Boundary discovered vector	146
Walking-one active sources	[146]
Walking-zero inactive sources	[146]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 125/168 — DI2[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI2[7]
FPGA package ball	AE6
FPGA boundary cell	351
Expected source vector	147
Measured CPLD channel	2
Measured CPLD pin	91
Gateware discovered vector	147
Boundary discovered vector	147
Walking-one active sources	[147]
Walking-zero inactive sources	[147]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 126/168 — DI3[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[0]
FPGA package ball	AA8
FPGA boundary cell	291
Expected source vector	126
Measured CPLD channel	3
Measured CPLD pin	99
Gateware discovered vector	126
Boundary discovered vector	126
Walking-one active sources	[126]
Walking-zero inactive sources	[126]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 127/168 — DI3[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[1]
FPGA package ball	AD11
FPGA boundary cell	225
Expected source vector	127
Measured CPLD channel	3
Measured CPLD pin	97
Gateware discovered vector	127
Boundary discovered vector	127
Walking-one active sources	[127]
Walking-zero inactive sources	[127]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 128/168 — DI3[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[2]
FPGA package ball	AE11
FPGA boundary cell	222
Expected source vector	128
Measured CPLD channel	3
Measured CPLD pin	96
Gateware discovered vector	128
Boundary discovered vector	128
Walking-one active sources	[128]
Walking-zero inactive sources	[128]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 129/168 — DI3[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[3]
FPGA package ball	V12
FPGA boundary cell	189
Expected source vector	129
Measured CPLD channel	3
Measured CPLD pin	95
Gateware discovered vector	129
Boundary discovered vector	129
Walking-one active sources	[129]
Walking-zero inactive sources	[129]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 130/168 — DI3[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[4]
FPGA package ball	Y12
FPGA boundary cell	228
Expected source vector	130
Measured CPLD channel	3
Measured CPLD pin	94
Gateware discovered vector	130
Boundary discovered vector	130
Walking-one active sources	[130]
Walking-zero inactive sources	[130]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 131/168 — DI3[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[5]
FPGA package ball	Y13
FPGA boundary cell	231
Expected source vector	131
Measured CPLD channel	3
Measured CPLD pin	93
Gateware discovered vector	131
Boundary discovered vector	131
Walking-one active sources	[131]
Walking-zero inactive sources	[131]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 132/168 — DI3[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[6]
FPGA package ball	AA12
FPGA boundary cell	240
Expected source vector	132
Measured CPLD channel	3
Measured CPLD pin	92
Gateware discovered vector	132
Boundary discovered vector	132
Walking-one active sources	[132]
Walking-zero inactive sources	[132]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 133/168 — DI3[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI3[7]
FPGA package ball	AA13
FPGA boundary cell	243
Expected source vector	133
Measured CPLD channel	3
Measured CPLD pin	91
Gateware discovered vector	133
Boundary discovered vector	133
Walking-one active sources	[133]
Walking-zero inactive sources	[133]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 134/168 — DI4[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[2]
FPGA package ball	AA19
FPGA boundary cell	93
Expected source vector	114
Measured CPLD channel	4
Measured CPLD pin	96
Gateware discovered vector	114
Boundary discovered vector	114
Walking-one active sources	[114]
Walking-zero inactive sources	[114]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 135/168 — DI4[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[3]
FPGA package ball	AC19
FPGA boundary cell	87
Expected source vector	115
Measured CPLD channel	4
Measured CPLD pin	95
Gateware discovered vector	115
Boundary discovered vector	115
Walking-one active sources	[115]
Walking-zero inactive sources	[115]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 136/168 — DI4[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[6]
FPGA package ball	AB20
FPGA boundary cell	78
Expected source vector	118
Measured CPLD channel	4
Measured CPLD pin	92
Gateware discovered vector	118
Boundary discovered vector	118
Walking-one active sources	[118]
Walking-zero inactive sources	[118]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 137/168 — DI4[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[8]
FPGA package ball	W19
FPGA boundary cell	60
Expected source vector	120
Measured CPLD channel	4
Measured CPLD pin	90
Gateware discovered vector	120
Boundary discovered vector	120
Walking-one active sources	[120]
Walking-zero inactive sources	[120]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 138/168 — DI5[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[10]
FPGA package ball	AE26
FPGA boundary cell	558
Expected source vector	108
Measured CPLD channel	5
Measured CPLD pin	87
Gateware discovered vector	108
Boundary discovered vector	108
Walking-one active sources	[108]
Walking-zero inactive sources	[108]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 139/168 — DI5[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI5[11]
FPGA package ball	AD21
FPGA boundary cell	573
Expected source vector	109
Measured CPLD channel	5
Measured CPLD pin	86
Gateware discovered vector	109
Boundary discovered vector	109
Walking-one active sources	[109]
Walking-zero inactive sources	[109]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 140/168 — DI6[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[0]
FPGA package ball	AD25
FPGA boundary cell	549
Expected source vector	84
Measured CPLD channel	6
Measured CPLD pin	99
Gateware discovered vector	84
Boundary discovered vector	84
Walking-one active sources	[84]
Walking-zero inactive sources	[84]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 141/168 — DI6[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[1]
FPGA package ball	AD26
FPGA boundary cell	561
Expected source vector	85
Measured CPLD channel	6
Measured CPLD pin	97
Gateware discovered vector	85
Boundary discovered vector	85
Walking-one active sources	[85]
Walking-zero inactive sources	[85]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 142/168 — DI6[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[2]
FPGA package ball	AC21
FPGA boundary cell	576
Expected source vector	86
Measured CPLD channel	6
Measured CPLD pin	96
Gateware discovered vector	86
Boundary discovered vector	86
Walking-one active sources	[86]
Walking-zero inactive sources	[86]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 143/168 — DI6[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[3]
FPGA package ball	AC22
FPGA boundary cell	582
Expected source vector	87
Measured CPLD channel	6
Measured CPLD pin	95
Gateware discovered vector	87
Boundary discovered vector	87
Walking-one active sources	[87]
Walking-zero inactive sources	[87]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 144/168 — DI6[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[4]
FPGA package ball	AC23
FPGA boundary cell	603
Expected source vector	88
Measured CPLD channel	6
Measured CPLD pin	94
Gateware discovered vector	88
Boundary discovered vector	88
Walking-one active sources	[88]
Walking-zero inactive sources	[88]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 145/168 — DI6[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[5]
FPGA package ball	AD24
FPGA boundary cell	588
Expected source vector	89
Measured CPLD channel	6
Measured CPLD pin	93
Gateware discovered vector	89
Boundary discovered vector	89
Walking-one active sources	[89]
Walking-zero inactive sources	[89]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 146/168 — DI6[9]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[9]
FPGA package ball	AB24
FPGA boundary cell	618
Expected source vector	93
Measured CPLD channel	6
Measured CPLD pin	89
Gateware discovered vector	93
Boundary discovered vector	93
Walking-one active sources	[93]
Walking-zero inactive sources	[93]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 147/168 — DI6[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[10]
FPGA package ball	AB25
FPGA boundary cell	642
Expected source vector	94
Measured CPLD channel	6
Measured CPLD pin	87
Gateware discovered vector	94
Boundary discovered vector	94
Walking-one active sources	[94]
Walking-zero inactive sources	[94]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 148/168 — DI6[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[11]
FPGA package ball	AB26
FPGA boundary cell	633
Expected source vector	95
Measured CPLD channel	6
Measured CPLD pin	86
Gateware discovered vector	95
Boundary discovered vector	95
Walking-one active sources	[95]
Walking-zero inactive sources	[95]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 149/168 — DI6[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[12]
FPGA package ball	AA22
FPGA boundary cell	606
Expected source vector	96
Measured CPLD channel	6
Measured CPLD pin	85
Gateware discovered vector	96
Boundary discovered vector	96
Walking-one active sources	[96]
Walking-zero inactive sources	[96]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 150/168 — DI7[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[3]
FPGA package ball	Y21
FPGA boundary cell	594
Expected source vector	73
Measured CPLD channel	7
Measured CPLD pin	95
Gateware discovered vector	73
Boundary discovered vector	73
Walking-one active sources	[73]
Walking-zero inactive sources	[73]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 151/168 — DI7[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[4]
FPGA package ball	Y22
FPGA boundary cell	609
Expected source vector	74
Measured CPLD channel	7
Measured CPLD pin	94
Gateware discovered vector	74
Boundary discovered vector	74
Walking-one active sources	[74]
Walking-zero inactive sources	[74]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 152/168 — DI7[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[5]
FPGA package ball	Y23
FPGA boundary cell	615
Expected source vector	75
Measured CPLD channel	7
Measured CPLD pin	93
Gateware discovered vector	75
Boundary discovered vector	75
Walking-one active sources	[75]
Walking-zero inactive sources	[75]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 153/168 — DI7[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[6]
FPGA package ball	Y25
FPGA boundary cell	627
Expected source vector	76
Measured CPLD channel	7
Measured CPLD pin	92
Gateware discovered vector	76
Boundary discovered vector	76
Walking-one active sources	[76]
Walking-zero inactive sources	[76]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 154/168 — DI7[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[7]
FPGA package ball	Y26
FPGA boundary cell	624
Expected source vector	77
Measured CPLD channel	7
Measured CPLD pin	91
Gateware discovered vector	77
Boundary discovered vector	77
Walking-one active sources	[77]
Walking-zero inactive sources	[77]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 155/168 — DI7[10]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[10]
FPGA package ball	W23
FPGA boundary cell	639
Expected source vector	80
Measured CPLD channel	7
Measured CPLD pin	87
Gateware discovered vector	80
Boundary discovered vector	80
Walking-one active sources	[80]
Walking-zero inactive sources	[80]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 156/168 — DI7[11]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[11]
FPGA package ball	W24
FPGA boundary cell	636
Expected source vector	81
Measured CPLD channel	7
Measured CPLD pin	86
Gateware discovered vector	81
Boundary discovered vector	81
Walking-one active sources	[81]
Walking-zero inactive sources	[81]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 157/168 — DI7[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI7[12]
FPGA package ball	W25
FPGA boundary cell	657
Expected source vector	82
Measured CPLD channel	7
Measured CPLD pin	85
Gateware discovered vector	82
Boundary discovered vector	82
Walking-one active sources	[82]
Walking-zero inactive sources	[82]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 158/168 — DI8[0]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[0]
FPGA package ball	V21
FPGA boundary cell	651
Expected source vector	56
Measured CPLD channel	8
Measured CPLD pin	99
Gateware discovered vector	56
Boundary discovered vector	56
Walking-one active sources	[56]
Walking-zero inactive sources	[56]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 159/168 — DI8[1]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[1]
FPGA package ball	V22
FPGA boundary cell	678
Expected source vector	57
Measured CPLD channel	8
Measured CPLD pin	97
Gateware discovered vector	57
Boundary discovered vector	57
Walking-one active sources	[57]
Walking-zero inactive sources	[57]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 160/168 — DI8[2]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[2]
FPGA package ball	V23
FPGA boundary cell	669
Expected source vector	58
Measured CPLD channel	8
Measured CPLD pin	96
Gateware discovered vector	58
Boundary discovered vector	58
Walking-one active sources	[58]
Walking-zero inactive sources	[58]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 161/168 — DI8[3]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[3]
FPGA package ball	V24
FPGA boundary cell	666
Expected source vector	59
Measured CPLD channel	8
Measured CPLD pin	95
Gateware discovered vector	59
Boundary discovered vector	59
Walking-one active sources	[59]
Walking-zero inactive sources	[59]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 162/168 — DI8[4]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[4]
FPGA package ball	V26
FPGA boundary cell	660
Expected source vector	60
Measured CPLD channel	8
Measured CPLD pin	94
Gateware discovered vector	60
Boundary discovered vector	60
Walking-one active sources	[60]
Walking-zero inactive sources	[60]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 163/168 — DI8[5]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[5]
FPGA package ball	U21
FPGA boundary cell	684
Expected source vector	61
Measured CPLD channel	8
Measured CPLD pin	93
Gateware discovered vector	61
Boundary discovered vector	61
Walking-one active sources	[61]
Walking-zero inactive sources	[61]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 164/168 — DI8[6]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[6]
FPGA package ball	U22
FPGA boundary cell	681
Expected source vector	62
Measured CPLD channel	8
Measured CPLD pin	92
Gateware discovered vector	62
Boundary discovered vector	62
Walking-one active sources	[62]
Walking-zero inactive sources	[62]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 165/168 — DI8[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[7]
FPGA package ball	U24
FPGA boundary cell	675
Expected source vector	63
Measured CPLD channel	8
Measured CPLD pin	91
Gateware discovered vector	63
Boundary discovered vector	63
Walking-one active sources	[63]
Walking-zero inactive sources	[63]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 166/168 — DI8[8]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI8[8]
FPGA package ball	U25
FPGA boundary cell	672
Expected source vector	64
Measured CPLD channel	8
Measured CPLD pin	90
Gateware discovered vector	64
Boundary discovered vector	64
Walking-one active sources	[64]
Walking-zero inactive sources	[64]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 167/168 — DI4[12]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI4[12]
FPGA package ball	AA20
FPGA boundary cell	90
Expected source vector	124
Measured CPLD channel	4
Measured CPLD pin	85
Gateware discovered vector	124
Boundary discovered vector	124
Walking-one active sources	[124]
Walking-zero inactive sources	[124]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

Link 168/168 — DI6[7]

Historical measured 338-vector discovery in each of two independent receive paths. Source logs are reduced original laboratory measurements; every required vector is present.

Check	Observed / expected
Net	DI6[7]
FPGA package ball	AC24
FPGA boundary cell	600
Expected source vector	91
Measured CPLD channel	6
Measured CPLD pin	91
Gateware discovered vector	91
Boundary discovered vector	91
Walking-one active sources	[91]
Walking-zero inactive sources	[91]
Gateware verdict	True
Boundary verdict	True
CPLD functional restoration	True

Procedure: preload sampled non-link boundary values, drive one CPLD link at a time, observe independent FPGA boundary cells and input capture, compare with schematic pin map; reset TAPs to release EXTEST.

Acceptance: exactly one discovered source, matching walking-one and walking-zero source indices and the schematic; both all-zero and all-one baselines correct; functional mode restored. Not an at-speed or analog qualification.

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

10 Per-channel ADC test cards

Physical channel 01 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 01 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 02 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 02 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 03 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 03 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 04 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 04 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 05 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 05 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 06 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 06 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 07 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 07 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 08 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 08 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 09 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 09 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 10 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 10 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 11 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 11 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog characterization NOT MEASURED. No CFD analog acceptance is inferred from synthetic transport or a free-running physical ADC buffer. Fit/route the simulator and collect this channel in a later batch.

Physical channel 12 — ADC A

Digital capture	Recorded result
Synthetic bank0	kind=synthetic transport validation; samples=4096
Physical bank0	kind=physical CPLD/ADC inputs

Analog line exercise	Observed
pass_	True
distinct_settings	257
interior_samples	241856
visited_codes	3129
stimulus_adequate	True

Bit	Zero samples	One samples	Result
0	120576	121280	PASS
1	120752	121104	PASS
2	120741	121115	PASS
3	120414	121442	PASS
4	121425	120431	PASS
5	120677	121179	PASS
6	121025	120831	PASS
7	121599	120257	PASS
8	120930	120926	PASS
9	120921	120935	PASS
10	121121	120735	PASS
11	121039	120817	PASS

Pair	00 / 01 / 10 / 11 counts	Result
[0, 1]	[58655, 62097, 61921, 59183]	PASS
[1, 2]	[60265, 60476, 60487, 60628]	PASS
[2, 3]	[58516, 61898, 62225, 59217]	PASS
[3, 4]	[60489, 60936, 59925, 60506]	PASS
[4, 5]	[60696, 59981, 60729, 60450]	PASS
[5, 6]	[60388, 60637, 60289, 60542]	PASS
[6, 7]	[60574, 61025, 60451, 59806]	PASS
[7, 8]	[60944, 59986, 60655, 60271]	PASS
[8, 9]	[60301, 60620, 60629, 60306]	PASS
[9, 10]	[60643, 60478, 60278, 60457]	PASS
[10, 11]	[60960, 60079, 60161, 60656]	PASS

Every displayed statistic was regenerated from the scan report and independently checked against retained raw STR-decoded buffers. Raw input hashes are in the evidence manifest. Missing finite-grid codes alone do not fail this test.

Physical channel 12 — ADC B

Digital capture	Recorded result
Synthetic bank1	kind=synthetic transport validation; samples=4096
Physical bank1	kind=physical CPLD/ADC inputs

Analog line exercise	Observed
pass_	True
distinct_settings	257
interior_samples	241760
visited_codes	3057
stimulus_adequate	True

Bit	Zero samples	One samples	Result
0	120764	120996	PASS
1	120865	120895	PASS
2	119582	122178	PASS
3	121041	120719	PASS
4	120011	121749	PASS
5	121959	119801	PASS
6	121197	120563	PASS
7	120249	121511	PASS
8	120270	121490	PASS
9	120854	120906	PASS
10	120655	121105	PASS
11	120976	120784	PASS

Pair	00 / 01 / 10 / 11 counts	Result
[0, 1]	[61388, 59477, 59376, 61519]	PASS
[1, 2]	[60945, 58637, 59920, 62258]	PASS
[2, 3]	[60285, 60756, 59297, 61422]	PASS
[3, 4]	[59991, 60020, 61050, 60699]	PASS
[4, 5]	[60178, 61781, 59833, 59968]	PASS
[5, 6]	[61183, 60014, 60776, 59787]	PASS
[6, 7]	[60731, 59518, 60466, 61045]	PASS
[7, 8]	[59860, 60410, 60389, 61101]	PASS
[8, 9]	[59961, 60893, 60309, 60597]	PASS
[9, 10]	[60300, 60355, 60554, 60551]	PASS
[10, 11]	[60979, 59997, 59676, 61108]	PASS

Every displayed statistic was regenerated from the scan report and independently checked against retained raw STR-decoded buffers. Raw input hashes are in the evidence manifest. Missing finite-grid codes alone do not fail this test.

11 Raw data, reproducibility and review

Batch	Mode / original status
00-historical	HISTORICAL / INCOMPLETE

The inputs directory retains complete evidence bundles. files.sha256.json lists every file hash and size. Generated figures retain their offline analysis JSON. Edit report.tex for typography and content.tex for this report; edit the source template for future reports. Rebuilding figures does not contact instruments.

Review checklist: verify factory UID and photographs; confirm pin-level cabling; approve numerical acceptance limits and calibration status; review every failure and deviation; confirm channel coverage and cleanup; retain raw records with this signed PDF.

No result is an assertion that the hardware is free of every possible defect. The evidence defines tested behavior, test conditions and observed limitations. This dossier is not a substitute for the separate CERN-equivalent detector validation bench.